

Limitations of Tail Recursion

Tail recursive functions require rearranging their code to make them suitable for the tail call optimization feature of the compiler or interpreter. However, not all compilers and interpreters offer tail call optimization as a feature. Notably, CPython (the Python interpreter downloaded from <https://python.org>) does not implement tail call optimization. Even if you write your recursive functions with the recursive call as the last action, it will still cause stack overflows after enough function calls are made.

Not only that, but CPython will likely never have tail call optimization as a feature. Guido van Rossum, the creator of the Python programming language, has explained that tail call optimization can make programs harder to debug. Tail call optimization removes frame objects from the call stack, which in turn removes the debugging information that frame objects can provide. He also points out that once tail call optimization is implemented, Python programmers will begin to write code that depends on the feature, and their code won't run on non-CPython interpreters that don't implement tail call optimization.

Finally, and I concur, van Rossum disagrees with the idea that recursion is a fundamentally important part of day-to-day programming. Computer scientists and mathematicians tend to place recursion on a pedestal. But tail call optimization is simply a workaround hack to make some recursive algorithms actually workable, rather than simply crashing with a stack overflow.

While CPython doesn't feature tail call optimization, this doesn't mean another compiler or interpreter that implements the Python language couldn't have tail call optimization. Unless tail call optimization is explicitly a part of a programming language specification, it is not a feature of a programming language, but rather of individual compilers or interpreters of a programming language.

The lack of tail call optimization is not unique to Python. The Java compiler since version 8 also doesn't support tail call optimization. Tail call optimization is a part of the ECMAScript 6 version of JavaScript; however, as of 2022, only the Safari web browser's implementation of JavaScript